THE GEOMETRY OF ATTENTION: RICCI CURVATURE AND TRANSFORMERS TRAINING AND ROBUSTNESS

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ABSTRACT

Transformer models have revolutionized machine learning, and the theoretical underpinnings behind their success are only now starting to be explored. In this work, we analyze the performance and robustness of transformers by considering the attention mechanism as a graph operator, focusing on the geometry of attention maps viewed as weighted graphs. Specifically, we investigate the role of Ricci curvature, a metric closely tied to graph spectral properties and system robustness, in shaping the training dynamics and robustness of transformers. Our theoretical analysis establishes a link between Ricci curvature and the convergence of gradient descent on transformers, and consequently, their training and fine tuning. We also show that a higher frequency of more positive values in the Ricci curvature distribution of attention graphs, therefore more system robustness, leads to more robust transformers, highlighting the impact of curvature on the robustness of transformers. Leveraging these insights, we propose an efficient regularization method to train curvature-adjusted transformers. Supporting our theoretical findings, experiments show that our proposed attention curvature manipulation can improve the learning speed, performance, or generalizability of vision and language transformers. Additionally, our observations point to a trade-off between their performance and robustness. This work demonstrates that the geometry of the attention map provides a theoretically elegant and computationally versatile framework for analyzing and manipulating transformers training, generalization, performance, and robustness, opening new avenues for designing models using geometric concepts.

1 INTRODUCTION

Transformers (Vaswani et al., 2017) have become the cornerstone of modern machine learning, excelling in tasks across natural language processing (Brown et al., 2020; Devlin et al., 2018) and computer vision (Dosovitskiy et al., 2020; Liu et al., 2021b; Touvron et al., 2021). Recent studies have begun to explain the inner workings of transformers, analyzing their expressive power (Yun et al., 2020; Zaheer et al., 2020), their generalization properties (Li et al., 2023; Zhou et al., 2024), in-context learning (Von Oswald et al., 2023), learning dynamics (Abbe et al., 2024; Dovonon et al., 2024), and robustness (Bhojanapalli et al., 2021; Shao et al., 2021). A rigorous understanding of how transformers process their input remains elusive, especially through mathematical tools offering practical solutions. To address this gap, we analyze transformers by treating attention mechanisms as graph operators and exploring for the first time the *geometry* of attention maps as weighted graphs. Focusing on graph Ricci curvature (Bauer et al., 2017; Ollivier, 2009), a well-established metric closely tied to graph spectral properties and system robustness (Bauer et al., 2011; Pouryahya et al., 2017), our findings reveal its connection to the training dynamics and robustness of transformers.

Our work reveals that the Ricci curvature of the attention graph plays a crucial role in shaping the training dynamics and robustness of transformers. By analyzing the geometry of attention graphs considering their spectral properties, we establish a direct link between Ricci curvature and the convergence behavior of gradient descent in transformers. Specifically, we demonstrate that Ricci curvature, with established links to system robustness, controls a proxy linked to the convergence rate during training, implied from our Lemma 4.2. Additionally, analyzing the statistical mechanics of transformers by conceptualizing the attention mechanism as a system of interacting particles, we uncover how the Ricci curvature distribution of the attention graph influences the robustness of transformers. Visually summarized in Figure 1, this dual perspective provides a comprehensive

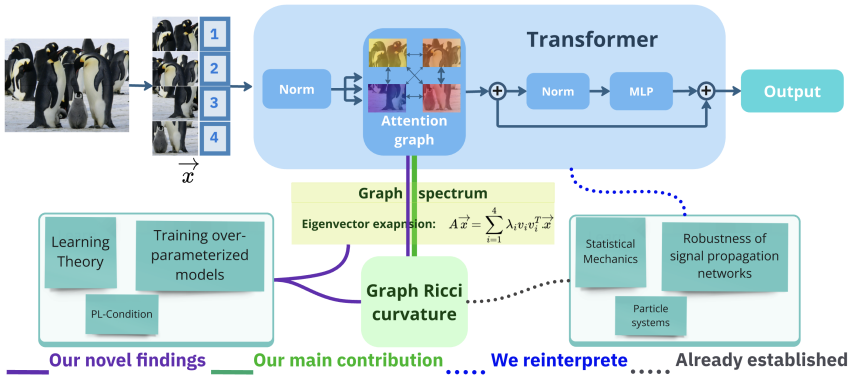

Figure 1: A visual summary of the main concepts and contributions. The lines between each block indicate the connection between concepts and tools that are already established in the literature (dotted black), where we provide complementary reinterpretation (dotted blue), our main contributions (solid green), or otherwise novel connections established in this paper (solid purple).

understanding of how the geometry of attention maps affects both the performance and robustness of transformer models. Informed by our theoretical and empirical findings, we then introduce a simple yet efficient training method that manipulates the curvature distribution of attention maps through a variance-adjusting regularization term. Our experiments show that increasing variance speeds up loss optimization, whereas decreasing it enhances performance at the cost of reduced generalizability. These experiments also illuminate the trade-off between performance and robustness, demonstrating that changes in the curvature can enhance one at the potential expense of the other.

Main contributions

- We establish a theoretical link between Ricci curvature in attention graphs and gradient descent convergence in transformers.
- Using a statistical mechanics perspective, we show how the Ricci curvature distribution in attention graphs affects transformer robustness, particularly against input perturbations.
- We introduce a training method that allows for curvature adjustment in the attention graph, influencing performance and generalizability of transformers, validated by empirical results.
- Linking attention graph geometry to key aspects of transformer behavior, this paper takes the first step toward applying geometric tools to study and improve the attention mechanism.

To recap, this paper marks a step toward elucidating the success of transformers by revealing how the geometry of attention graphs influences their behavior. Through a combination of theoretical insights, empirical investigations, and practical methods, we demonstrate how understanding and manipulating this geometry can enhance training, performance, and robustness. To our knowledge, this is the first work to connect the geometry of attention to these critical aspects of transformers, offering new avenues for designing more effective transformer models, through geometric insights and techniques.

2 Background

Transformers and attention mechanism. The core component of transformers is the attention mechanism, which allows the model to account for dependencies between input tokens (Vaswani et al., 2017). Given an input $X = (x_1, \ldots, x_n)$, where $x_i \in \mathbb{R}^d$ is the embedding of the i-th token, for the value matrix $V \in \mathbb{R}^{n \times d_v}$, the attention scores are computed based on pairwise similarities between the query ($Q \in \mathbb{R}^{n \times d_k}$) and key ($K \in \mathbb{R}^{n \times d_k}$) representations,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \tag{1}$$

The attention matrix $A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)$ can be viewed as the adjacency matrix of a graph, where nodes correspond to tokens and edge weights to attention scores. Focusing on an architecture common to the most widely adopted transformer models, in this paper we consider the self-attention mechanism in a weighted fully-connected graph (full attention), where $Q = XW_Q$, $K = XW_K$, and $V = XW_V$ with learnable weight matrices $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$.

Graph Ricci curvature. Ricci curvature (Figure 2) captures the extent to which the local geometry induced by a Riemannian metric deviates from that of a Euclidean space (Bauer et al., 2017). Graph Ricci curvature is a powerful tool in deep learning (Di Giovanni et al., 2023; Liu et al., 2023; Southern et al., 2023; Topping et al., 2021). We use the Ollivier-Ricci curvature (ORC) (Ollivier, 2009), an optimal transport formulation of graph Ricci curvature, defined as

$$\kappa_{OR}(v, u) := 1 - \frac{W_1(\mu_v, \mu_u)}{d_G(v, u)}, \tag{2}$$

for an edge $(v, u) \in \mathcal{E}$ on a graph $G = (\mathcal{V}, \mathcal{E})$, where μ_v and μ_u are probability measure on the nodes anchoring (v, u), $d_G(.)$ represents a distance metric on $\mathcal{V}$, and W_1 denotes the 1-Wasserstein distance (Lin et al., 2011; Jost & Liu, 2014).

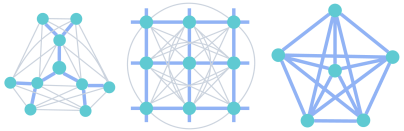

Figure 2: Three weighted complete graphs with highlighted larger weight edges (in blue). Different edge weight distributions result in negative (left), near zero (middle), and positive (right) Ricci curvature. Curvature in a weighted graph is negative/zero/positive when removing low-weight edges (in gray) yields a tree/grid/complete graph, respectively.

Ricci curvature and system robustness. Ricci curvature has been linked to system robustness (Pouryahya et al., 2017), owing to its connection with entropy. This established link is highlighted by the following fundamental result from optimal transport (Lott & Villani, 2009), which bounds entropy based on a lower bound of the Ricci curvature (Pouryahya et al., 2017),

$$S(\mu_\lambda) \geq (1 - \lambda)\, S(\mu_0) + \lambda S(\mu_1) + \underline{k}\frac{\lambda(1 - \lambda)}{2} W_2\left(\mu_0, \mu_1\right)^2, \tag{3}$$

where $S(\cdot)$ denotes the Boltzmann entropy (Adkins, 1983), $\underline{k}$ bounds the Ricci curvature from below, $W_2\left(\mu_0, \mu_1\right)$ is the Wasserstein distance of order 2 between μ_0 and μ_1 in the metric space $(P(\mathcal{X}), W_2)$ of probability measures on $\mathcal{X}$, and μ_λ is a measure on the geodesic between μ_0 and μ_1 given by $\lambda \in [0, 1]$. This inequality suggests that Ricci curvature and entropy are positively correlated (Pouryahya et al., 2017). Furthermore, there exists a correlation between *system robustness* and entropy. System robustness refers to the ability of a system to quickly return to its stationary state following a perturbation (Demetrius & Manke, 2005).The Fluctuation Theorem (Evans et al., 1993) establishes a positive correlation between the robustness and entropy of a system, implying that system robustness and curvature are positively correlated (Pouryahya et al., 2017).

3 Graph Geometry of Learning Representations in Transformers

Attention mechanisms in transformer models can be viewed as graph operators, where the tokens serve as nodes and the attention scores as edge weights in a complete graph. The geometric properties of this attention graph inherently shape the representation space, with the eigenvalues and eigenvectors dictating the direction and scale of transformations applied to each input. This, in turn, determines the geometry of the representations that the transformer learns. Our analysis demonstrates that Ricci curvature values in the attention graph are linked to the convergence rate during training as well as the robustness of transformers to input perturbations.

Informally stating our main results, in this section, we explain how viewing the attention mechanism as a graph allows us to reach these findings. We adopt two complementary perspectives, combining a mathematical one that connects our graph-based analysis to the training dynamics of transformers, and a physics-based one that interprets the transformer as a system of interacting particles. The mathematical perspective links the ORC of the attention graph to the convergence of gradient descent-based optimization, while the physics perspective explores the statistical mechanics of transformers, drawing insights from the interplay between graph geometry, dynamics, and robustness in networked